

Marmara University, 2020

Modeling and Discrete Simulation

Subject 7

Random-Variate Generation

Mujdat Soyturk, Ph.D.
Associate Professor

Purpose & Overview



- Develop understanding of generating samples from a specified distribution as input to a simulation model.
- Illustrate some widely-used techniques for generating random variates.
  - Inverse-transform technique
  - Acceptance-rejection technique
  - Special properties

Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

Contents

- Inverse-transform technique
- Acceptance-rejection technique
- Special properties

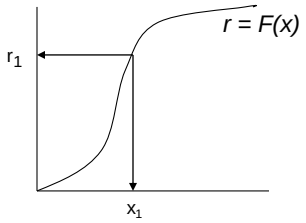
Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

INVERSE-TRANSFORM

Inverse-transform Technique

- The concept:
  - For *cdf* function: $r = F(x)$
  - Generate r from uniform $(0,1)$
  - Find x:

$$x = F^{-1}(r)$$



Inverse-transform Technique

4 steps in the inverse-transform technique:

- Compute the ***cdf*** of the desired random variable X.
- Set $\boldsymbol{F(X) = R}$ on the range of X.
- Solve the equation $\boldsymbol{F(X) = R}$ for X in terms of $\boldsymbol{R}$.
- Generate uniform random numbers $R_1, R_2, R_3, \ldots$ and compute the desired ***random variates*** by

$$X_i = F^{-1}(R_i)$$

Exponential Distribution

- Exponential Distribution:
  - Exponential *cdf*:

$r = F(x)$
$= 1 - e^{-\lambda x}$ for $x \geq 0$

  - To generate $X_1, X_2, X_3 \ldots$

$X_i = F^{-1}(R_i)$
$= -(1/\lambda)\ln(1-R_i)$

[Random-Variate Generator]

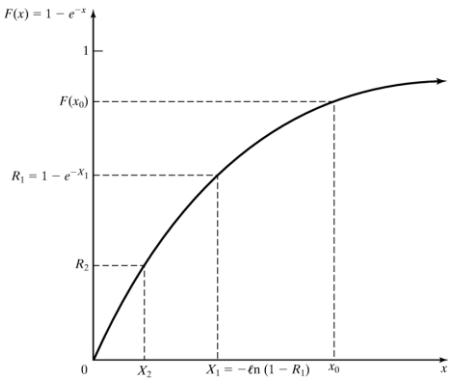


Figure: Inverse-transform technique for *exp(λ = 1)*

Exponential Distribution

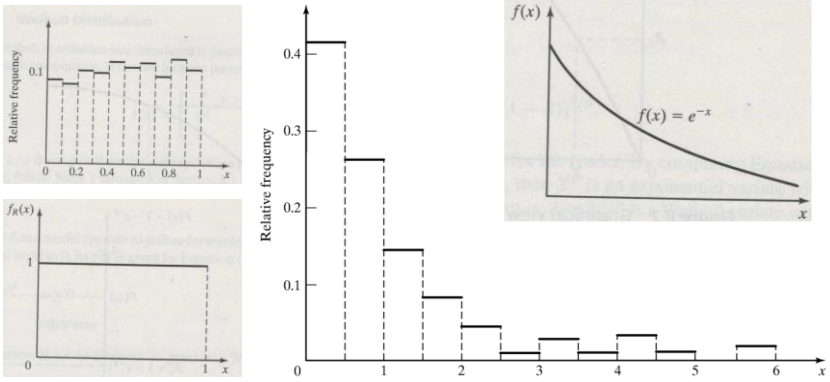
[Random-Variate Generator]

$X_i = F^{-1}(R_i)$
$= -(1/\lambda)\ln(1-R_i)$

i	1	2	3	4	5
R_i	0.1306	0.0422	0.6597	0.7965	0.7696
X_i	0.1400	0.0431	1.078	1.592	1.468

Exponential Distribution

- Example: Generate *200* variates X_i with distribution *exp(λ= 1)*
  - Generate *200* ***R****s* with *U(0,1)* and utilize *Random-Variate Generator*, the histogram of ***X***s become:



Other Distributions

- Examples of other distributions for which **inverse *cdf*** works are:
  - Uniform distribution
  - Weibull distribution
  - Triangular distribution

Empirical Continuous Dist'n

- When theoretical distribution is not applicable
- To collect empirical data:
  - Resample the observed data
  - Interpolate between observed data points to fill in the gaps
- For a small sample set (size *n*):
  - Arrange the data from smallest to largest

$$x_{(1)} \le x_{(2)} \le \ldots \le x_{(n)}$$

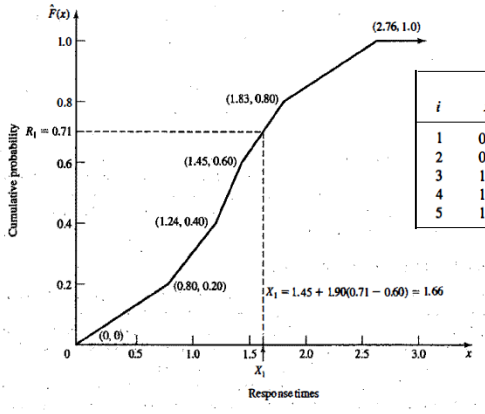
  - Assign the probability 1/n to each interval $x_{(i-1)} \le x \le x_{(i)}$

$$X = \hat{F}^{-1}(R) = x_{(i-1)} + a_i\left(R - \frac{(i-1)}{n}\right)$$

where $a_i = \frac{x_{(i)} - x_{(i-1)}}{i/n - (i-1)/n} = \frac{x_{(i)} - x_{(i-1)}}{1/n}$

Empirical Continuous Dist'n

2.76, 1.83, 0.80, 1.45, 1.24 ➔ Sorted sequence: 0.80, 1.24, 1.45, 1.83, 2.76



i	*Interval* $x_{(i-1)} < x \le x_{(i)}$	*Probability* 1/*n*	*Cumulative Probability, i/n*	*Slope* a_i
1	$0.0 < x \le 0.80$	0.2	0.2	4.00
2	$0.80 < x \le 1.24$	0.2	0.4	2.20
3	$1.24 < x \le 1.45$	0.2	0.6	1.05
4	$1.45 < x \le 1.83$	0.2	0.8	1.90
5	$1.83 < x \le 2.76$	0.2	1.0	4.65

$$a_1 = \frac{x_1 - x_0}{\frac{i}{n} - \frac{i-1}{n}} = \frac{0.8 - 0}{0.2 - 0} = 4$$

$$X = x_{(i-1)} + a_i\left(R - \frac{(i-1)}{n}\right)$$

Empirical Continuous Dist'n

If a large sample of data is available, it might be more convenient to summarize the data into a frequency distribution.

- c_i is the cumulative probability of the first i intervals of the frequency distributions
- $x_{(i-1)} < x < x_{(i)}$ is the ith interval
- Then the slope of the ith line segment is given by:

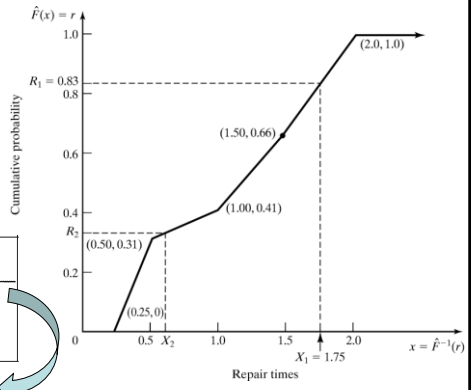
$$a_i = \frac{x_{(i)} - x_{(i-1)}}{c_i - c_{i-1}}$$

- And the inverse *cdf* is calculated as:

$$X = \hat{F}^{-1}(R) = x_{(i-1)} + a_i(R - c_{i-1})$$

i	Interval (Hours)	Frequency	Relative Frequency	Cumulative Frequency, c_i	Slope a_i
1	$0.25 \le x \le 0.5$	31	0.31	0.31	0.81
2	$0.5 < x \le 1.0$	10	0.10	0.41	5.0
3	$1.0 < x \le 1.5$	25	0.25	0.66	2.0
4	$1.5 < x \le 2.0$	34	0.34	1.00	1.47

$$a_1 = \frac{x_1 - x_0}{c_1 - c_0} = \frac{0.25 - 0}{0.31 - 0} = 0.81$$



Empirical Continuous Dist'n

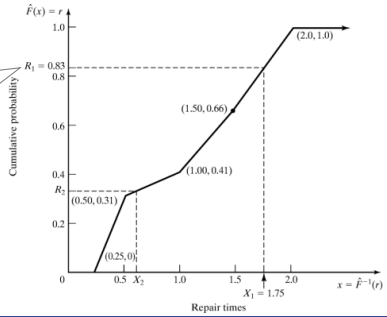
Example: Suppose the data collected for 100 broken-widget repair times are:

i	Interval (Hours)	Frequency	Relative Frequency	Cumulative Frequency, c_i	Slope, a_i
1	$0.25 \le x \le 0.5$	31	0.31	0.31	0.81
2	$0.5 \le x \le 1.0$	10	0.10	0.41	5.0
3	$1.0 \le x \le 1.5$	25	0.25	0.66	2.0
4	$1.5 \le x \le 2.0$	34	0.34	1.00	1.47

Consider $R_1 = 0.83$:

$c_3 = 0.66 < R_1 < c_4 = 1.00$

$X_1 = x_{(4-1)} + a_4(R_1 - c_{(4-1)})$
$= 1.5 + 1.47(0.83-0.66)$
$= 1.75$



Discrete Distribution

- All discrete distributions can be generated via inverse-transform technique
- Method: numerically, through a table-lookup procedure, or algebraically, in terms of a formula
- Examples of application:
  - Empirical
  - Discrete uniform
  - Gamma

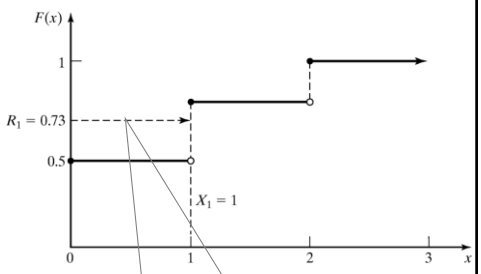
Discrete Distribution

- Example: Suppose the number of shipments, *x*, on the loading dock of IHW company is either *0*, *1*, or *2*
  - Data - Probability distribution:

x	*p(x)*	*F(x)*
0	0.50	0.50
1	0.30	0.80
2	0.20	1.00

  - Method - **Given R, the generation scheme becomes:**

$$x = \begin{cases} 0, & R \le 0.5 \\ 1, & 0.5 \prec R \le 0.8 \\ 2, & 0.8 \prec R \le 1.0 \end{cases}$$



Consider $R_1 = 0.73$:
$F(x_{i-1}) < R <= F(x_i)$
$F(x_0) < 0.73 <= F(x_1)$
Hence, $x_1 = 1$

ACCEPTANCE-REJECTION

Acceptance-Rejection technique

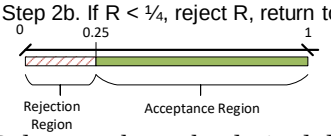
- Useful particularly when inverse *cdf* does not exist in closed form, a.k.a. thinning
- Illustration: To generate random variates, $X \sim U(1/4, 1)$

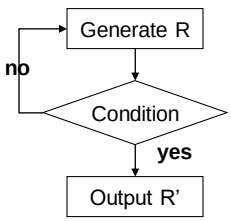
Procedures:

Step 1. Generate R ~ U[0,1]

Step 2a. If R >= ¼, accept X=R.

Step 2b. If R < ¼, reject R, return to Step 1



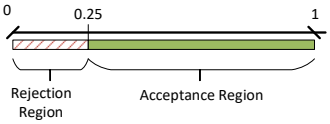


- R does not have the desired distribution, but R conditioned (R') on the event $\{R \geq ¼\}$ does.
- Efficiency: Depends heavily on the ability to minimize the number of rejections. ➔ what is the efficiency for this example?

Acceptance-Rejection technique

- Efficiency:
  - What is the mean number of random numbers $\boldsymbol{R}$ required to generate one variate $\boldsymbol{X}$?
    - R ranges from 0 to 1.
    - R' ranges from .25 to 1.

    ➔4/3



To generate 1000 values of $\boldsymbol{X}$ would require appr. 1333 random numbers $\boldsymbol{R}$.

Alternative approach:

$$X = \frac{1}{4} + \left(\frac{3}{4}R\right)$$

Acceptance-Rejection technique

- **Poisson Distribution**

A Poisson random variable, N, with mean $\alpha > 0$ has pmf

$$p(n) = P(N = n) = \frac{e^{-\alpha}\alpha^n}{n!}, \quad n = 0, 1, 2, \ldots$$

Arrivals to the system (interarrivals): ***N = n*, if and only if**

$$A_1 + A_2 + \cdots + A_n \le 1 < A_1 + \cdots + A_n + A_{n+1}$$

$$A_i = (-1/\alpha) \ln R_i$$

$$\sum_{i=1}^{n} -\frac{1}{\alpha} \ln R_i \le 1 < \sum_{i=1}^{n+1} -\frac{1}{\alpha} \ln R_i$$

$$\ln \prod_{i=1}^{n} R_i = \sum_{i=1}^{n} \ln R_i \ge -\alpha > \sum_{i=1}^{n+1} \ln R_i = \ln \prod_{i=1}^{n+1} R_i$$

$$\prod_{i=1}^{n} R_i \ge e^{-\alpha} > \prod_{i=1}^{n+1} R_i$$

Acceptance-Rejection technique

- **Poisson Distribution**
  - Random variate generation

Step 1. Set $n = 0$, $P = 1$.

Step 2. Generate a random number R_{n+1}, and replace P by $P \cdot R_{n+1}$.

Step 3. If $P < e^{-\alpha}$, then accept $N = n$. Otherwise, reject the current n, increase n by one, and return to step 2.

Acceptance-Rejection technique

- **Poisson Distribution**
  - Random variate generation

Generate three Poisson variates with mean $\alpha = 0.2$. First, compute $e^{-\alpha} = e^{-0.2} = 0.8187$. Next, get a sequence of random numbers R from Table A.1 and follow the previously described Steps 1 to 3:

Step 1. Set $n = 0$, $P = 1$.

Step 2. $R_1 = 0.4357$, $P = 1 \cdot R_1 = 0.4357$.

Step 3. Since $P = 0.4357 < e^{-\alpha} = 0.8187$, accept $N = 0$.

Step 1–3. ($R_1 = 0.4146$ leads to $N = 0$.)

Step 1. Set $n = 0$, $P = 1$.

Step 2. $R_1 = 0.8353$, $P = 1 \cdot R_1 = 0.8353$.

Step 3. Since $P \geq e^{-\alpha}$, reject $n = 0$ and return to Step 2 with $n = 1$.

Step 2. $R_2 = 0.9952$, $P = R_1 R_2 = 0.8313$.

Step 3. Since $P \geq e^{-\alpha}$, reject $n = 1$ and return to Step 2 with $n = 2$.

Step 2. $R_3 = 0.8004$, $P = R_1 R_2 R_3 = 0.6654$.

Step 3. Since $P < e^{-\alpha}$, accept $N = 2$.

n	R_{n+1}	P	*Accept/Reject*	*Result*
0	0.4357	0.4357	$P < e^{-\alpha}$ (accept)	$N = 0$
0	0.4146	0.4146	$P < e^{-\alpha}$ (accept)	$N = 0$
0	0.8353	0.8353	$P \geq e^{-\alpha}$ (reject)	
1	0.9952	0.8313	$P \geq e^{-\alpha}$ (reject)	
2	0.8004	0.6654	$P < e^{-\alpha}$ (accept)	$N = 2$

Acceptance-Rejection technique

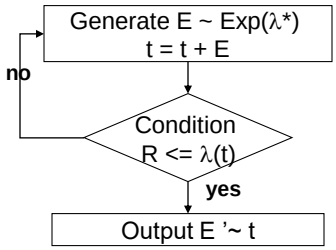
- **Poisson Distribution**
  - Efficiency:

How many random numbers will be required, on the average, to generate one Possion variate, *N*?

$$E(N+1) = \alpha + 1, \qquad \text{where } \boldsymbol{\alpha} \text{ is the mean.}$$

NSPP

- Non-stationary Poisson Process (NSPP): a Possion arrival process with an arrival rate that varies with time
- Idea behind thinning:
  - Generate a stationary Poisson arrival process at the fastest rate, $\lambda^* = \max \lambda(t)$
  - But "accept" only a portion of arrivals, thinning out just enough to get the desired time-varying rate



NSPP

- Example: Generate a random variate for a NSPP

Data: Arrival Rates

t (min)	*Mean Time Between Arrivals (min)*	*Arrival Rate λ(t) (#/min)*
0	15	1/15
60	12	1/12
120	7	1/7
180	5	1/5
240	8	1/8
300	10	1/10
360	15	1/15
420	20	1/20
480	20	1/20

Procedures:

Step 1. $\lambda^* = max\ \lambda_{(t)} = 1/5$, $t = 0$ and $i = 1$.

Step 2. For random number $R = 0.2130$,

$E = -5\ln(0.213) = 13.13$

$t = 13.13$

Step 3. Generate $R = 0.8830$

$\lambda_{(13.13)}/\lambda^* = (1/15)/(1/5) = 1/3$

Since $R > 1/3$, do not generate the arrival

Step 2. For random number $R = 0.5530$,

$E = -5\ln(0.553) = 2.96$

$t = 13.13 + 2.96 = 16.09$

Step 3. Generate $R = 0.0240$

$\lambda_{(16.09)}/\lambda^* = (1/15)/(1/5) = 1/3$

Since $R < 1/3$, $T_1 = t = 16.09$,

and $i = i + 1 = 2$

SPECIAL PROPERTIES

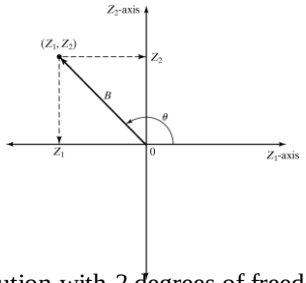
Special Properties

- Based on features of particular family of probability distributions
- For example:
  - Direct Transformation for normal and lognormal distributions
  - Convolution
  - Beta distribution (from gamma distribution)

Direct Transformation

- Approach for normal*(0,1)*:
  - Consider two standard normal random variables, Z_1 and Z_2, plotted as a point in the plane:

In polar coordinates:
$Z_1 = B \cos \phi$
$Z_2 = B \sin \phi$



  - $B^2 = Z^2_1 + Z^2_2 \sim$ chi-square distribution with *2* degrees of freedom = *Exp(λ = 2)*. Hence, $B = (-2 \ln R)^{1/2}$
  - The radius B and angle ϕ are mutually independent.

$$Z_1 = (-2 \ln R_1)^{1/2} \cos(2\pi R_2)$$
$$Z_2 = (-2 \ln R_1)^{1/2} \sin(2\pi R_2)$$

Direct Transformation

- Approach for normal(μ,σ^2):
  - Generate $Z_i \sim N(0,1)$

$$X_i = \mu + \sigma Z_i$$

- Approach for lognormal(μ,σ^2):
  - Generate $X \sim N((\mu,\sigma^2)$

$$Y_i = e^{X_i}$$

Summary

- Principles of random-variate generate via
  - Inverse-transform technique
  - Acceptance-rejection technique
  - Special properties
- Important for generating continuous and discrete distributions